Pricing an Optimal Carbon Credit for a Competitive Italian Biochar Supply Chain

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Motivation

- The most recent UN assessment no longer asks *whether* we cross 1.5 °C, but **how far past it** we go and **for how long**. Exceedance is widely assessed as unavoidable.

  UNEP, *Emissions Gap Report 2025: Off Target.*

- For **Italy** this is no joke:

  up to −6% of GDP

  The potential impact of global warming on Italy's GDP by mid-century, as estimated by the Euro-Mediterranean Center on Climate Change (CMCC).

  CMCC, Deloitte & EUI, via Euronews, 17 August 2026.

- **Biochar** emerges as a potentially extraordinary **ally against climate change**, thanks to its **carbon capture and storage** properties and its many possible practical applications.

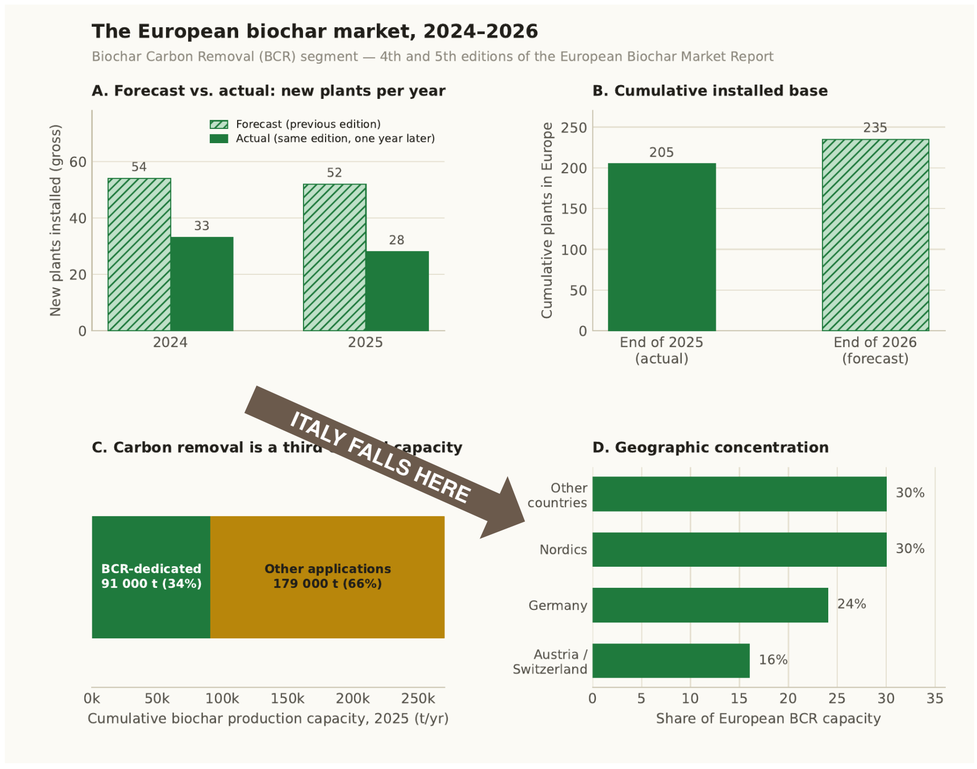
Can Italy develop a supply chain able to exploit the biochar solution?

⇒ *Is there an industrial demand for biochar in Italy?*

⇒ ***Can it be made competitive without spoiling its carbon-capture capability? Is there enough biomass?***

The Data Problem

- The European biochar market grows, but **below forecasts**: 33 new plants against 54 forecast in 2024, 28 against 52 in 2025.
- ∼70% of Europe's biochar carbon removal (BCR) capacity sits in **Germany**, the **Nordics** and **Austria/Switzerland**; **Italy** falls in the residual 'other countries'.



- There are **no reliable, publicly accessible databases** of measured data, either on biochar **prices** or on the **waste biomass** actually available across the territory. Every national source is a top-down aggregate or a modelled potential: none measures what a territory can actually mobilise.

Source	What it gives	Why it falls short
ENEA — Biomass Atlas	Potential of agro-industrial residues, at national scale	**Never completed**, due to funding cuts
JRC — EU Biomass Flows	Harmonised EU biomass production and trade, aggregated in flows	Categories **indistinguishable**; no domestic-vs-import split
S2Biom — Italy roadmap	Technical potential 2030: 34.4 Mt/yr of dry matter	A modelled 2030 projection, **theoretical**: not sited, not measured
CRRA — Annex A	The **most recent** national study; aggregates the above	**Internal inconsistencies** in the biochar data, which we **found and reported** to the authors

Picking Up the Phone: A First Step Towards Solving the Data Problem

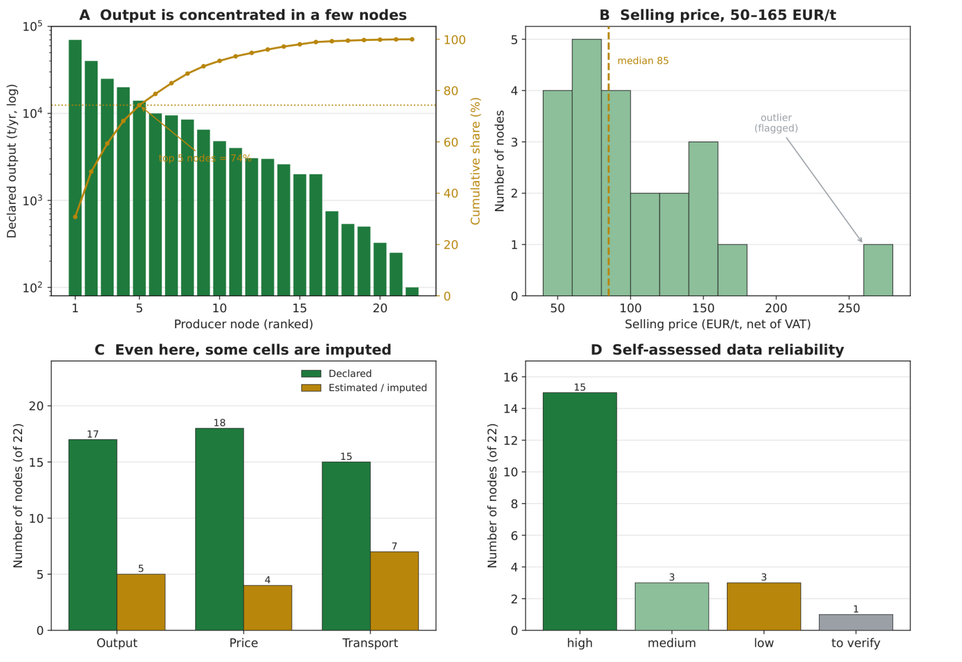
We called **131 wood-chip producers** one by one: **32** gave usable numbers and became geolocated nodes, reduced to the **22** that enter the model. On the biochar side:

6 biochar producers active in Italy

CREA 2023 survey (9 respondents).

An industrial demand for a carbon-black substitute exists

Pirelli asked for ∼7.6 t/day of biochar: the feedstock is not the problem, a competitive price is.



The anonymised wood-chip producer dataset. **(A)** output is concentrated: the top five nodes supply 74%; **(B)** selling prices span 50–165 €/t (median 85), with one flagged outlier; **(C)** even among the nodes that answered, many figures had to be estimated rather than declared; **(D)** each node's self-assessed data reliability.

Is there enough biomass? Yes. The 22 nodes could yield nearly **29,000 t of biochar per year** (about 79 t/day): roughly **ten times** the 7.6 t/day requested.

The research question

At what price should a carbon credit be set so that an optimised Italian biochar supply chain covers its costs while selling biochar at a price competitive with carbon black?

The Model

A mixed-integer linear program (MILP) for the siting of biochar plants, adapted to the Italian case from Grimm *et al.*

Inputs and outputs

- **Inputs:** the biomass nodes $j \in J$, each supplying S_j tonnes per year; the engineering specifications of the plants; the distances d_{ij} between nodes; the cost of building a new plant; the transport cost τ of moving biomass, per tonne-kilometre.
- **Outputs:** the optimised **geography** of the supply chain, $x_j \in \{0,1\}$ (1 = open plant, 0 = closed), and the **capacity** y_j of each open plant.
- Then: the **carbon credit** that makes costs and revenues break even.

Objective: minimise fixed plus transport cost

$$\min_{x,y,z} \sum_j C_j + \sum_{i,j} \tau\, d_{ij}\, S_i\, z_{ij}$$

where C_j is the annual fixed cost of plant j, following the six-tenths rule:

$$C_j = \underbrace{(\mathrm{CRF} + f_{\mathrm{OPEX}})}_{\text{annualised fixed and maintenance costs}} K_{\mathrm{ref}} \overbrace{\left(\frac{y_j}{y_{\mathrm{ref}}}\right)^{0.6}}^{\text{plant capacity}} \underbrace{x_j}_{\substack{1=\text{open}\\0=\text{closed}}}$$

$\mathrm{CRF} = r(1+r)^n/[(1+r)^n - 1]$: capital recovery factor (r discount rate, n plant lifetime); $K_{\mathrm{ref}}, y_{\mathrm{ref}}$: reference capital cost and capacity. A plant k times larger costs only $k^{0.6}$ times as much: **economies of scale**. The 0.6 power and the product with x_j are handled by piecewise linearisation.

Constraints

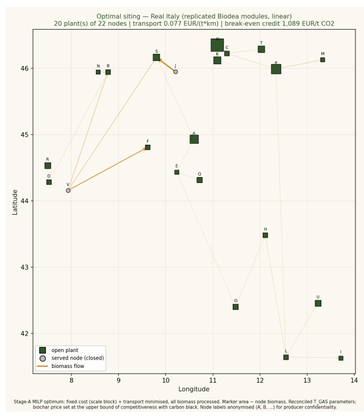
$z_{ij} \in [0,1]$: fraction of the biomass of node $i \in J$ processed at site $j \in J$; H: operating hours per year.

$\sum_j z_{ij} = 1 \quad \forall i$	every node uses all its biomass
$z_{ij} \le x_j \quad \forall i,j$	biomass goes only to open plants
$\sum_i S_i\, z_{ij} \le H\, y_j \quad \forall j$	a plant processes at most its capacity
$x_j \in \{0,1\},\ \ 0 \le z_{ij} \le 1$	

V. Grimm, K. Niazmand, P. Runge, *Renew. Sustain. Energy Rev.* **226**, 116298 (2026).

Results for the Northern Italian Supply Chain

The model is run on the 22-node network in two regimes: without and with economies of scale.



No economies of scale

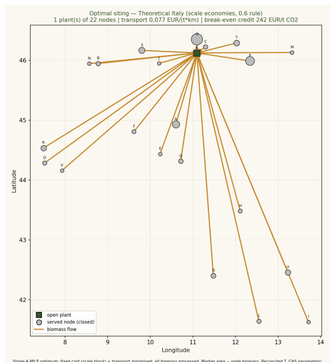
Identical small commercial modules stacked in parallel: cost grows linearly. The optimiser still finds non-obvious optima, switching off small nodes and hauling their biomass to saturate a module before opening a new one.

With economies of scale

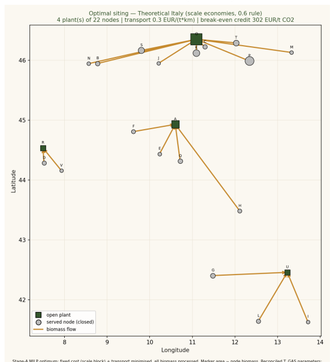
As transport gets dearer, the optimum breaks from a single central plant into several (maps below): the geography decentralises and the break-even credit rises.

Node positions are perturbed and labels anonymised for producer confidentiality.

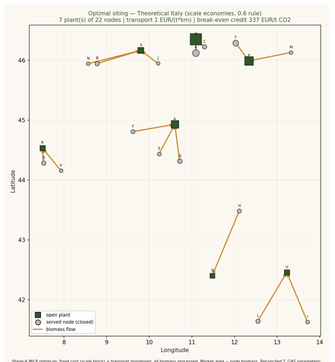
$\tau = 0.077$ **€/(t·km)**



$\tau = 0.3$ **€/(t·km)**



$\tau = 1$ **€/(t·km)**



Scenario	Open plants	Break-even credit
Economies of scale, $\tau = 0.077$ €/(t·km)	1	**242 €/tCO_2**
Economies of scale, $\tau = 0.3$ €/(t·km)	4	**302 €/tCO_2**
Economies of scale, $\tau = 1$ €/(t·km)	7	**337 €/tCO_2**
No economies of scale (modules in parallel)	20 sites	**1,089 €/tCO_2**

Life-Cycle Assessment (LCA)

The denominator of the credit is the net CO_2 actually removed, computed **following the Puro.earth methodology**: carbon durably stored in the biochar, minus the emissions the chain causes in biomass supply, production and end use.

$$\mathrm{CO_2^{seq}} = \rho_{\mathrm{bc}}\, B\, \chi_C\, \chi_{\mathrm{stab}}\, \tfrac{44}{12}$$

$$\mathrm{CO_2^{em}} = \underbrace{\mathrm{TK}\, e_{\mathrm{tr}}}_{\text{supply}} + \underbrace{B\, e_{\mathrm{pr}} + E\, e_{\mathrm{grid}} + K^{\mathrm{fix}} e_{\mathrm{cap}}}_{\text{production}} + \underbrace{\rho_{\mathrm{bc}} B\, (e_{\mathrm{use}} + \bar{d}\, e_{\mathrm{tr}})}_{\text{end use}}$$

$$\mathrm{CO_2^{net}} = \mathrm{CO_2^{seq}} - \mathrm{CO_2^{em}}$$

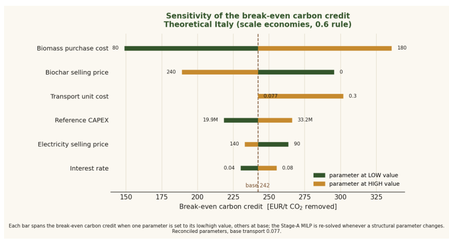
τ [€/(t·km)]	Net CO_2 removed [tCO_2/yr]
0.077	65,340
0.3	67,004
1	67,759

A more decentralised chain hauls biomass over shorter distances: lower transport emissions, more CO_2 removed.

ρ_{bc}: biochar yield; B: biomass processed; χ_C: carbon fraction; $\chi_{\mathrm{stab}} = 0.80$: stable fraction over 100 years (conservative end of the European Biochar Certificate range); TK: tonne-kilometres, the same aggregate charged in the transport cost; E: electricity produced; $K^{\mathrm{fix}} = \sum_j C_j$: annual fixed cost of the plants; $\bar{d}$: mean distance from plant to end use; $e_\bullet$: emission factors.

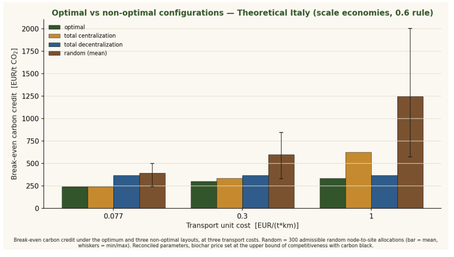
All net-removal values are overestimated by the model.

Limitations and Outlook



Each bar spans the break-even carbon credit when one parameter is set to its low/high value, others at base; the Stage-A MILP is re-solved whenever a structural parameter changes. Reconciled parameters, base transport 0.077.

Sensitivity (with economies of scale). The credit is dominated by the **biomass purchase price**, which swamps every other parameter: the spatial optimisation works, but is a **secondary lever**.



Break-even carbon credit under the optimum and three non-optimal layouts, at three transport costs. Random = 300 admissible random node-to-site allocations (bar = mean, whiskers = min/max). Reconciled parameters, biochar price set at the upper bound of competitiveness with carbon black.

Does optimisation matter? Where economies of scale can be exploited, the optimised chain is never worse than the extreme layouts (full centralisation or decentralisation) and far below random ones, for fixed parameter values.

Outlook: assumptions to relax

ASSUMPTION TO RELAX		PROPOSED IMPROVEMENT
Transport flat tariff on straight-line distances	⇒	Road-specific prices, optimising over the **real Italian road network**
Technology a single conversion technology (gasification)	⇒	Optimising over **different plant types** (pyrolysis, gasification, . . .)
Nodes plants only at producer nodes	⇒	Allowing **intermediate nodes** as candidate plant sites

A Complexity Scientist in the Biochar World

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